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F78AB
Actuarial and Financial Mathematics

Chapter 10: Discounted cash flows

LEADERS IN IDEAS AND SOLUTIONS

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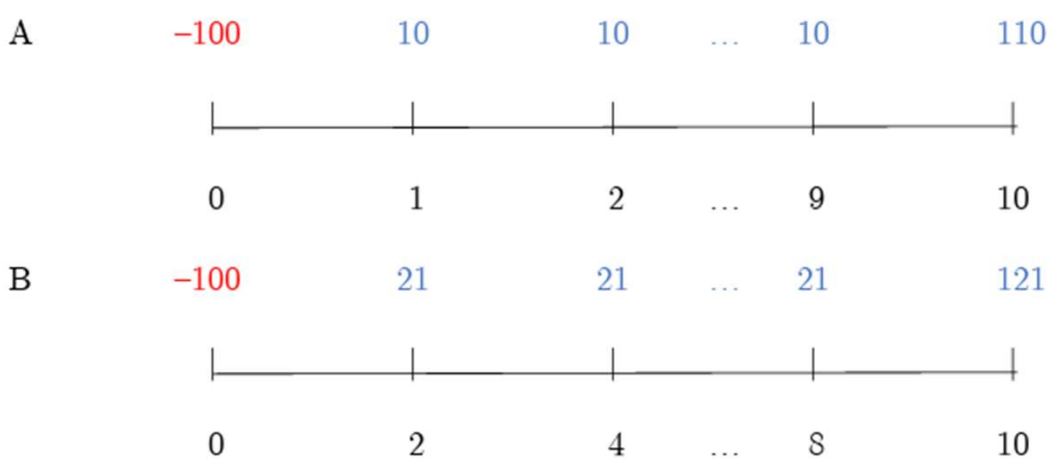
Recall Principle of Equivalence

We assume that any person or entity is indifferent about which of two equivalent series of cashflows they own. That is, they would be willing to exchange one set of cashflows for any other equivalent set of cashflows.

We use this principle to compare investments

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Example: Consider 2 investments, A and B, below



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Example continued

- The yield on both investments A and B is 10% per unit time effective
- **Check this as an exercise**

- If we could invest our money into the bank at 10%, then:
  - We would pay £100 for either investment; and
  - We would be indifferent between A and B

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Example continued

However, if we could find a bank account which paid 15% per time unit effective, we would consider the accumulated values of:

1) £100 invested directly into the bank for 10 years
2) Cashflows from A invested at 15%
3) Cashflows from B invested at 15%

And conclude:

- We would not pay £100 for either A or B
- We would prefer to invest in A than B

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Net Present Values (NPV)

- **Definition** Given a set of cashflows $c_1, c_2,\ldots,c_n$, at times $t_1, t_2,\ldots,t_n$ their net present value (NPV) is defined as:

$$\mathrm{NPV} = \sum_{r=0}^{n} c_r (1+i)^{-t_r}$$

- NPV is a function of *i* so we write NPV(*i*)
- Comparing NPVs of different investments at the market rate of interest allows us to choose between investments (Principle of Equivalence)
- When NPV(*i*) = 0, we obtain the yield, and we call this the Internal Rate of Return (IRR)

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Example

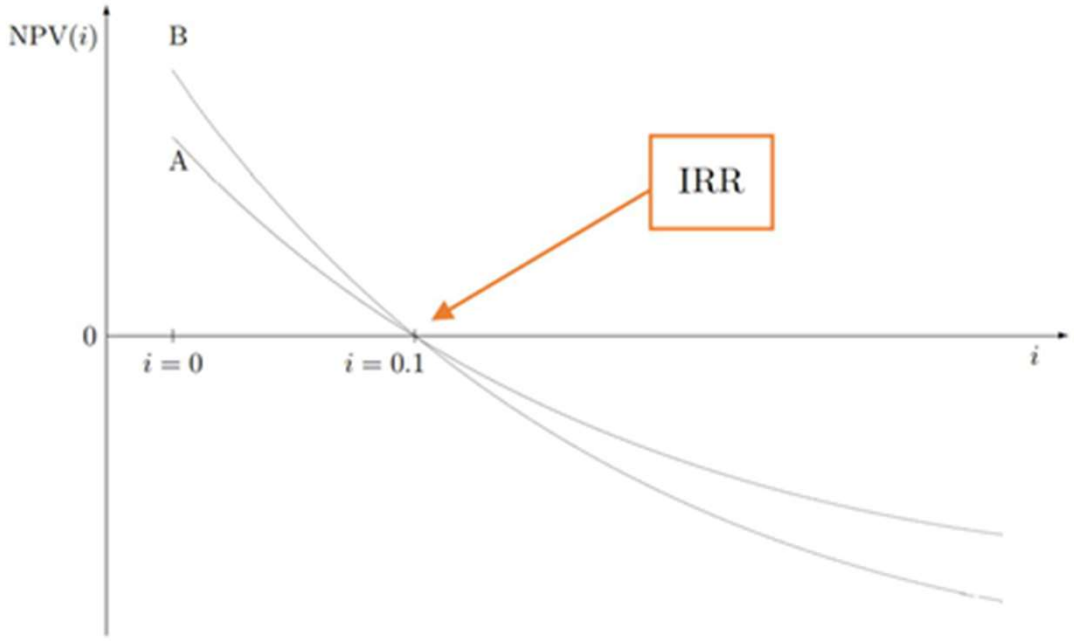
Consider the NPV functions of Investment A and B earlier:

$$\text{A (at time } t=0\text{): NPV}(i) = -100 + 10a_{\overline{10}|} + 100v^{10} = -100 + 10\left(\frac{1-v^{10}}{i}\right) + 100v^{10}$$

$$\text{B (at time } t=0\text{): NPV}(i) = -100 + 21\frac{a_{\overline{10}|}}{s_{\overline{2}|}} + 100v^{10} = -100 + 21\left(\frac{1-v^{10}}{(1+i)^2-1}\right) + 100v^{10}$$

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Graphically…



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Notes:

1. The yield exists if and only if the equation NPV(i) = 0 has a unique solution not less than -1.
2. The NPV function always exists and is informative even if the yield does not exist. See lecture notes for NPV functions of two investments that have no yield but can still be meaningfully compared
3, NPVs are additive – we could combine the cashflows of our two investments A and B to create a new investment, A+B, then:

$$\mathrm{NPV}_{\mathbf{A+B}}(i) = \mathrm{NPV}_{\mathbf{A}}(i) + \mathrm{NPV}_{\mathbf{B}}(i).$$

Example

Which of investments A or B would you prefer if the market rate of interest was:

(i) 5% p.a. effective?

(ii) 20% p.a. effective?

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Discounted Payback Period (DPP)

- Most investments involve an initial outlay, followed by receipt of income
- NPV calculates the profit from the investment
- If an investment has a positive NPV (at some interest rate *i*), then there must be a first time when the initial outlay has been recovered, allowing for interest
- That time is the **Discount Payback Period (DPP)** (or break-even point) calculated at interest rate *i.*

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Discounted Payback Period (DPP)

- Suppose we have a set of cashflows c_1, c_2,…,c_n, at times t_1, t_2,…,t_n , such that all negative cashflows precede all positive cashflows (ie initial outlays are followed by income) and an effective rate of interest *i* per time unit such that NPV*(i)* > 0.
- The Discounted Payback Period (DPP) at an effective rate of interest *i* per time unit is the smallest *k*, such that

$$\sum_{r=0}^{k} c_r v^{-t_r} \geq 0 .$$

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DPP example

You invest £100 in a five-year project, and receive cashflows of £30 at the end of each year. Given the market interest rate of 5%p.a. effective, find NPV(0.5) and the DPP.

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Payment schedule to determine DPP

- Think of the investment as a ‘loan’ paid back by each year’s profit. For the above example we can see that the loan has been repaid by end of year 4, so DPP is 4 (as we determined)

Time (Yrs)	‘Repayment’	‘Interest Content’	‘Capital Content’	‘Capital Outstanding’
0				100.00
1	30.00	5.00	25.00	75.00
2	30.00	3.75	26.25	48.75
3	30.00	2.44	27.56	21.19
4	30.00	1.06	28.94	−7.75
5	30.00	−0.39	30.39	−38.14

Accumulated profit

- Complex investment decisions can be evaluated by working out the net cashflows and calculating the discounted or accumulated profit.
- Decisions might include how much money is needed to finance the project, where to borrow the money needed, and the way in which that money must be repaid.

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Accumulated profit

- The accumulated profit is the net accumulated value at the end of the investment:

$$\mathrm{AP}\ (i) = \mathrm{NAV}(i) = \sum_{r=0}^{n} c_r (1+i)^{n-t_r}\ .$$

- Notice that

$$\mathrm{NPV} \times (1+i)^n = \mathrm{NAV} \text{ and } \mathrm{NAV} \times v^n = \mathrm{NPV}.$$

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Accumulated profit Example 1

Suppose you can both borrow and invest money at 6% p.a. effective. You borrow £100million, to be repaid by level annual repayments over ten years.

You use the money to finance a project that will make a profit of £18m at the end of each of the second to tenth years inclusive.

(i) Find the net cashflows

(ii) Calculate the accumulated profit at time $t = 10$

(iii) Calculate NPV (0.06)

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Comment on above example

1. This method is called the Discounted Cashflow (DCF) method – make direct use of cashflows by allowing for several transactions at once between different parties
2. There can be many parties involved in a complicated project
3. Each cashflow is considered separately, rather than reducing everything to discounted or projected values (where we lose information) – important because in practice we cannot borrow and invest money at the same effective rates of interest.

Accumulated profit Example 2

You wish to borrow £100 to invest in a project that will return profits of £18 at the end of each of the next 10 years. You borrow money from the bank which charges interest of 8% p.a. effective. However if you deposit or invest money you can only earn 6% p.a. effective interest. You have two possible ways to repay the loan:

(i) By net level payments of £14.90 (check this!) payable in arrears

(ii) By using the entire £18 p.a. profit to repay the loan as soon as possible, investing the profits to earn 6% interest.

Which method would you prefer?

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Accumulated profit Example 3

An investor wishes to build a factory costing £100,000,000 to make computer games consoles. It is expected to produce profits of £50,000,000 at the end of each of the second, third, fourth and fifth years, after which it will be written off.

The project can be financed in one of two ways:

a) Capital can be borrowed at 7% p.a. effective interest. The profits will be used to pay off the loan as soon as possible, and once that is done, further profits can be invested to earn 5%p.a. effective interest

b) The investor can issue a five-year bond with annual coupons of 6%, payable yearly in arrears. The bond will be issued at par, but must be redeemed at 110%. The investor can borrow money at 7%p.a. effective, while any profits can be invested to earn 5% p.a. effective.

Which method of funding should the investor choose?

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Attendance matters!

